PARSHWANATH CHARITABLE TRUST'S
A.P. SHAH INSTITUTE OF TECHNOLOGY
Department of Computer Science and Engineering
Data Science

BITTRUTH-To Verify Authenticity of Media

Shrikant Thakur (23107045)
Arjun Talekar (23107005)
Aayush Thakur (23107015)
Pranav Parab (23107049)

Project Guide
Ms.Swati Parhad

Outline

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- System Design
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Introduction

- **BitTruth** is a machine learning-based deepfake detection system designed to verify the authenticity of media.
- The system analyzes images and videos to detect facial inconsistencies and signs of manipulation.
- It classifies media as "Real" or "Fake" and provides a confidence score for the verification.

Motivation

- The primary driver is to reduce misinformation by flagging suspected deepfake content.
- It aims to help users verify images and videos before sharing them, thereby limiting the spread of fake news.

Objectives

- **Automated Authentication:** To develop a machine learning-based system capable of analyzing images and videos to detect facial inconsistencies and manipulation signs.
- **Media Classification:** To accurately classify digital media as "Real" or "Fake" while providing a specific confidence score for authenticity verification.
- **Information Integrity:** To reduce the spread of misinformation and fake news by flagging suspected deepfake content before it is shared.
- **Accessibility:** To design a simple and user-friendly interface that allows non-technical users to easily verify media authenticity.

Literature Survey of the existing system

Sr. No	Title	Author	Year	Demerits	Methodology	Outcome
1	Deepfake Detection Challenge (DFDC)	AWS, Facebook, Microsof	2020	High computational cost for real-time video processing.	Uses large-scale datasets of manipulated videos.	Established a baseline for identifying deepfakes using deep learning.

Literature Survey of the existing system

Sr. No	Title	Author	Year	Demerits	Methodology	Outcome
2	FaceForensics++: Learning to Detect Manipulated Facial Images	Rössler et al.	2019	Difficulty in detecting advanced or "unseen" deepfakes.	Automated benchmark using CNNs to extract facial features.	Proven effective at classifying media as Real or Fake.

Literature Survey of the existing system

Sr. No	Title	Author	Year	Demerits	Methodology	Outcome
3	Vision Transformer for Deepfake Detection	Research community	2021	Requires extensive hardware (GPU) for efficient training.	Uses Transformer models to analyze global image context.	Improved accuracy in detecting subtle facial inconsistencies.

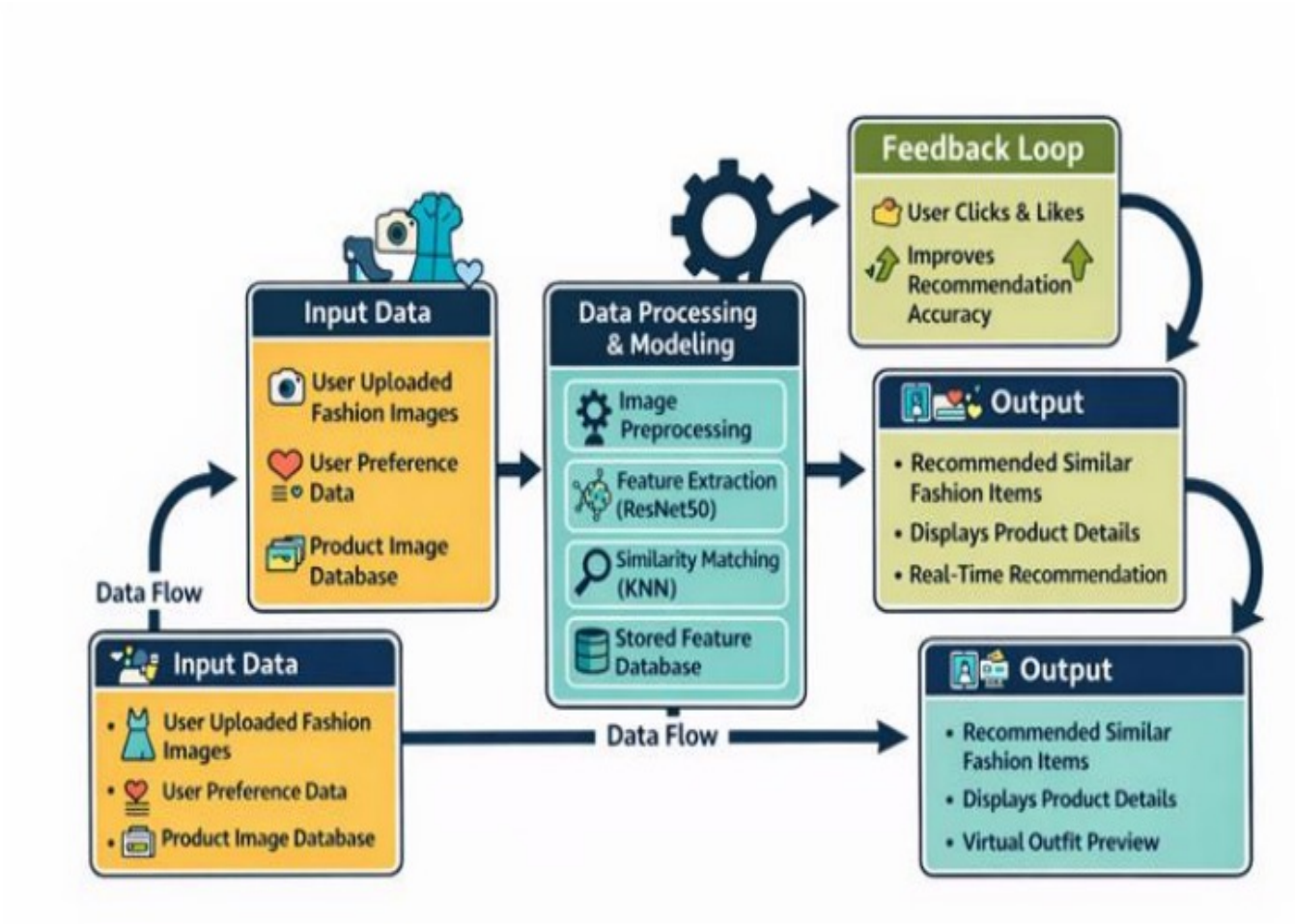
Limitations of existing systems

- **Lack of Diverse Data:** Current systems often struggle with biased results due to a shortage of varied datasets.
- **High Computational Cost:** Analyzing high-definition media typically requires expensive, GPU-enabled hardware.
- **Detection Gaps:** Advanced or "unseen" deepfakes frequently bypass older models, leading to false predictions.
- **Real-Time Lag:** Handling and processing large volumes of video data in real-time remains a major technical hurdle.
- **User Trust Issues:** Many systems lack transparency, making it difficult for non-technical users to understand or trust the results.

Problem statement

- Verifying the authenticity of media using only manual observation is often unreliable and does not give accurate results.
- Existing media platforms lack smart AI-based systems to detect sophisticated deepfakes and manipulated content.
- Current systems do not effectively analyze images and videos for facial inconsistencies and subtle signs of digital manipulation.
- There is a need for an AI-ML based system that allows automated authenticity verification and provides accurate confidence scores to reduce misinformation.

System Design



Technologies

- **Frontend Development:**
  - Html
  - Css
  - Java Script
- **Backend Development:**
  - Python
- **Machine Learning:**
  - **Frameworks:** TensorFlow / PyTorch
  - **Libraries:** OpenCV, NumPy, Pandas
  - **Models:** XceptionNet, EfficientNet-B4, CNN–LSTM, Vision Transformer
  - **Datasets:** FaceForensics++, CelebDF, DFDC, WildDeepfake

Methodologies

•Data Collection:

Compiled diverse media datasets including FaceForensics++, CelebDF, DFDC, and WildDeepfake.

•Data Preprocessing:

Implemented frame-based analysis to manage large video data and optimize processing speed.

Analyzed images and videos to detect facial inconsistencies and signs of digital manipulation.

•Model Selection:

Utilized deep learning architectures such as XceptionNet, EfficientNet-B4, and Vision Transformers.

Integrated CNN–LSTM models to evaluate temporal sequences in video content.

•Model Training and Evaluation:

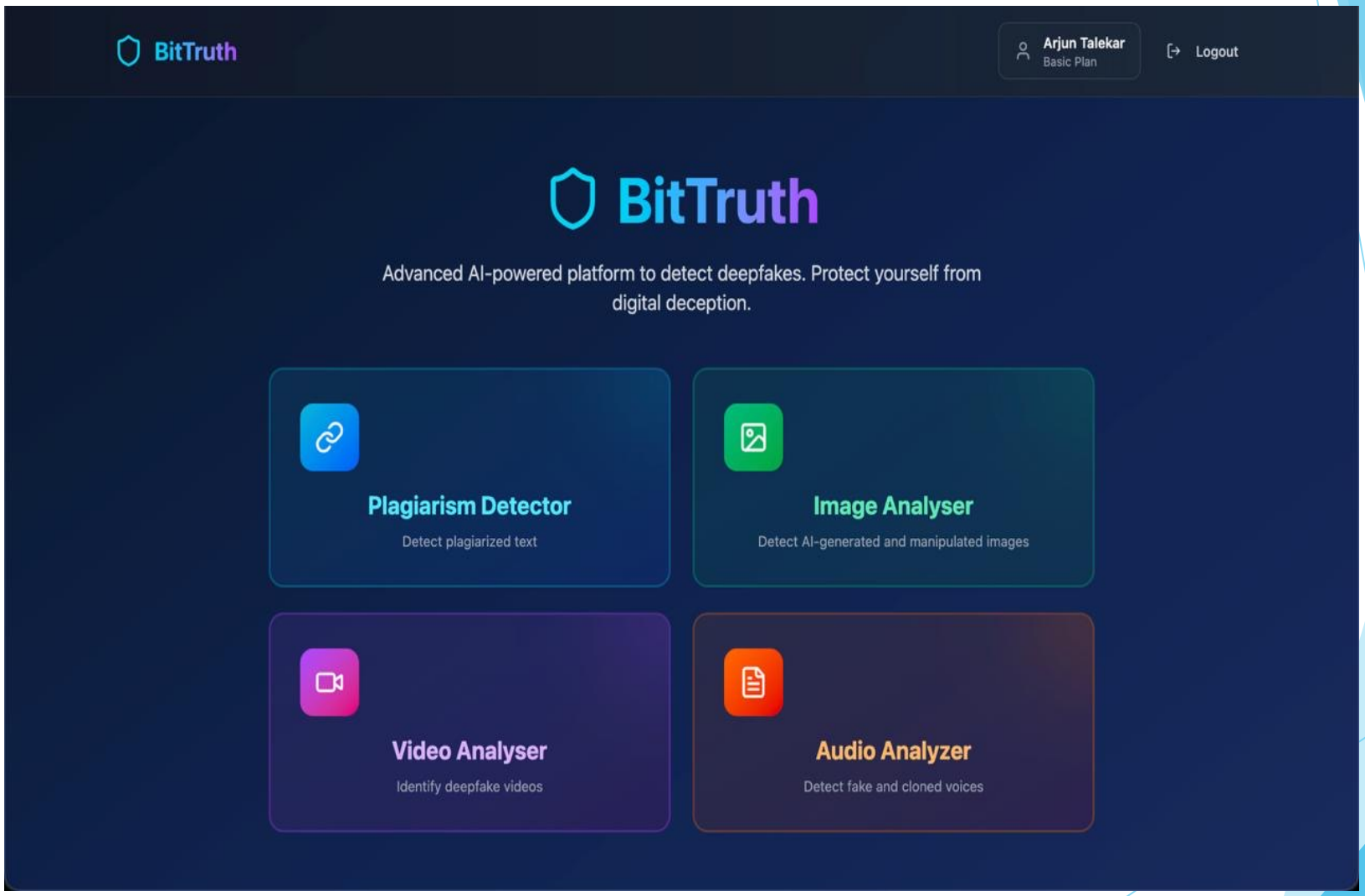
Developed a machine learning system to classify media as "Real" or "Fake".

Evaluated performance using public datasets to improve reliability and reduce false predictions.

•Prediction:

Provided a specific confidence score and final verdict to verify the authenticity of uploaded media.

Implementation



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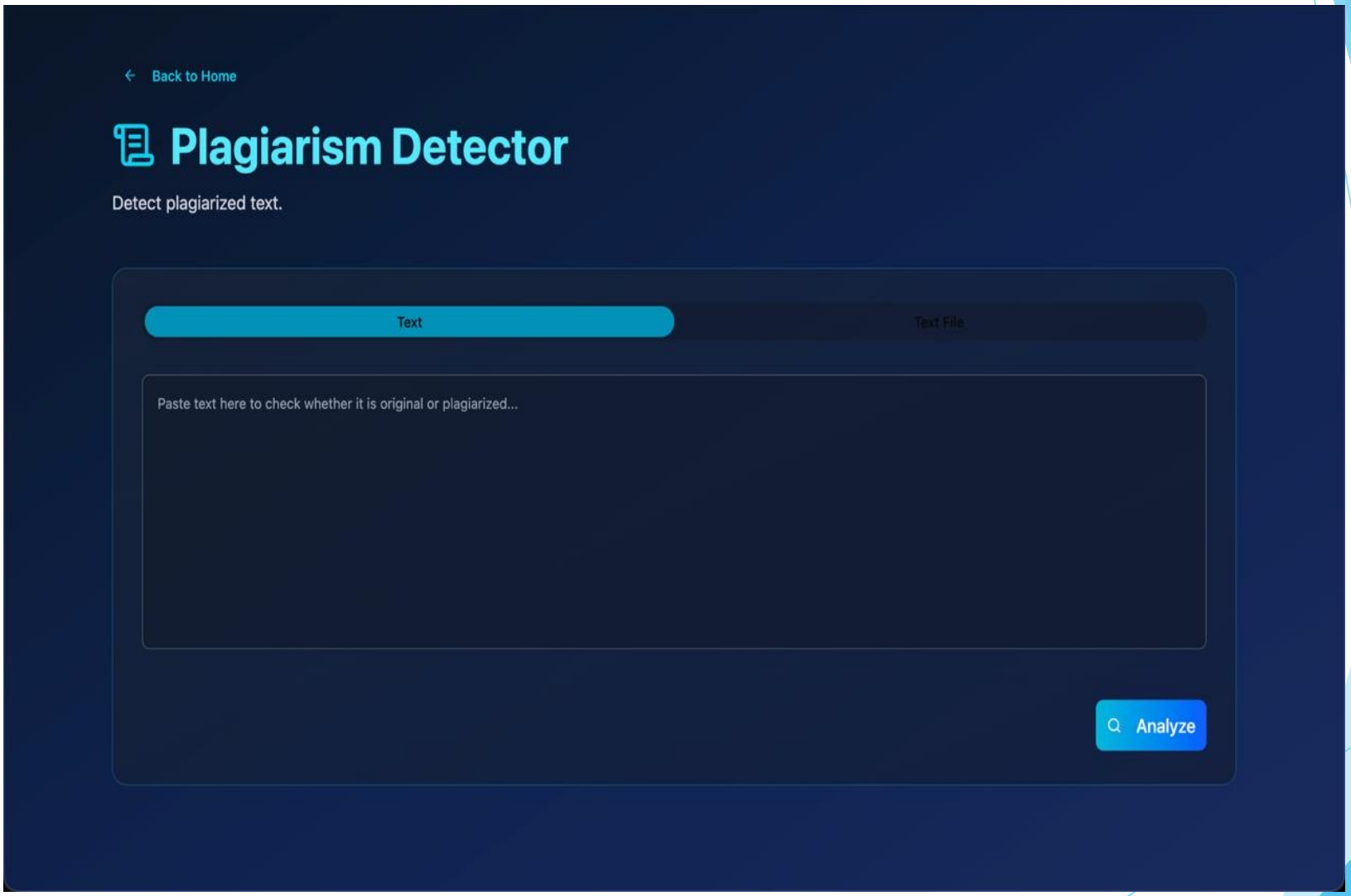

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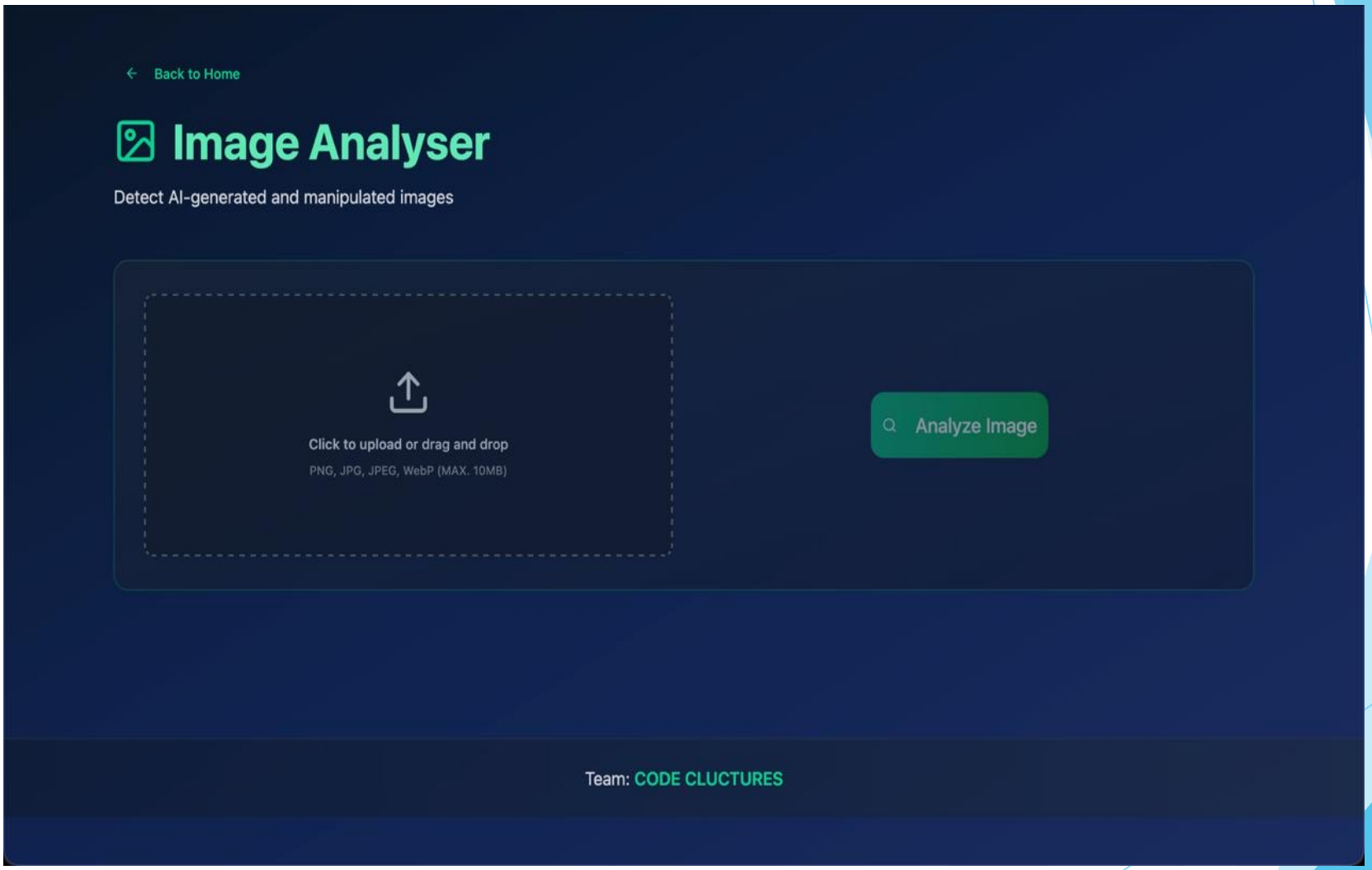

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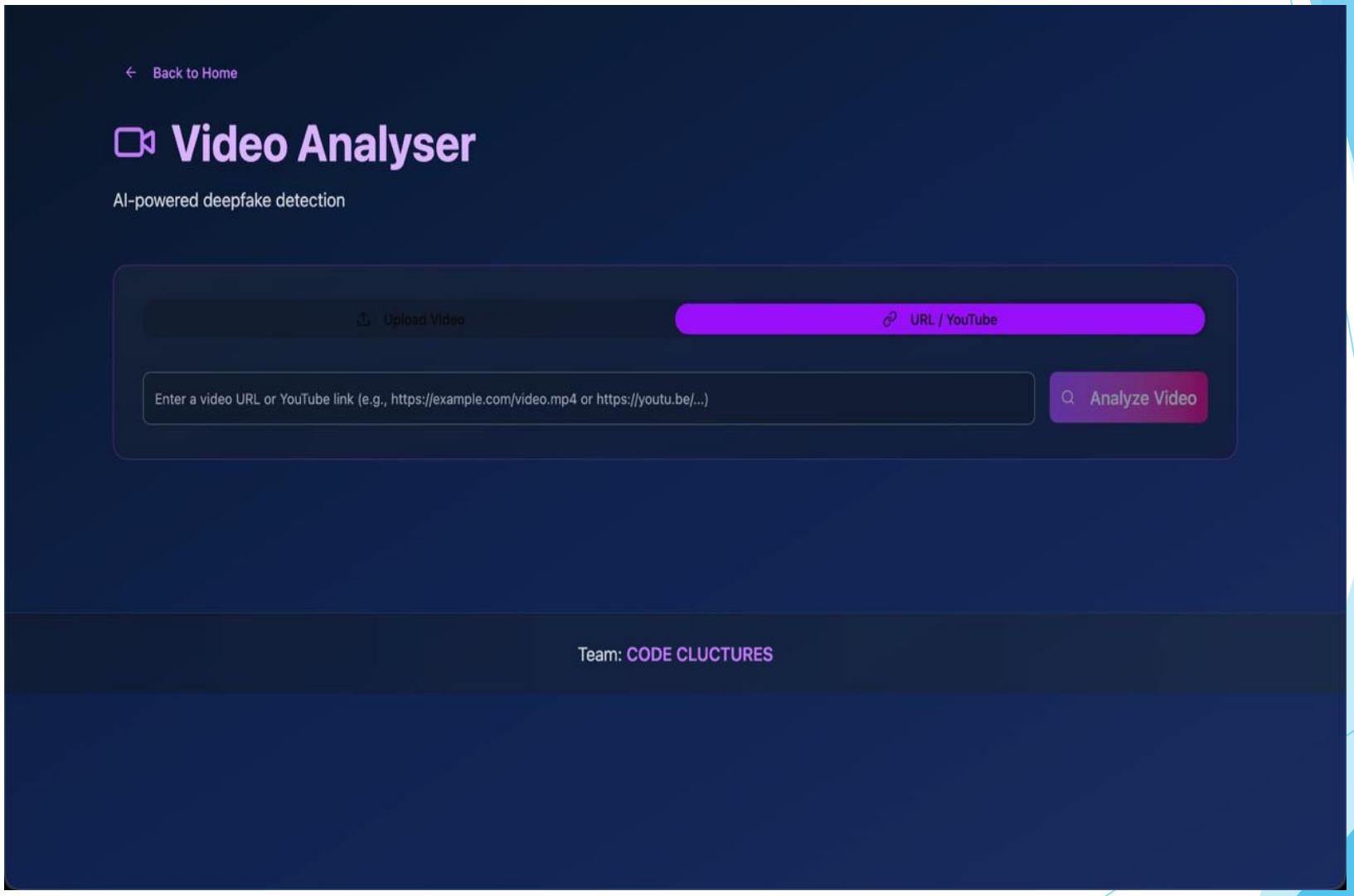

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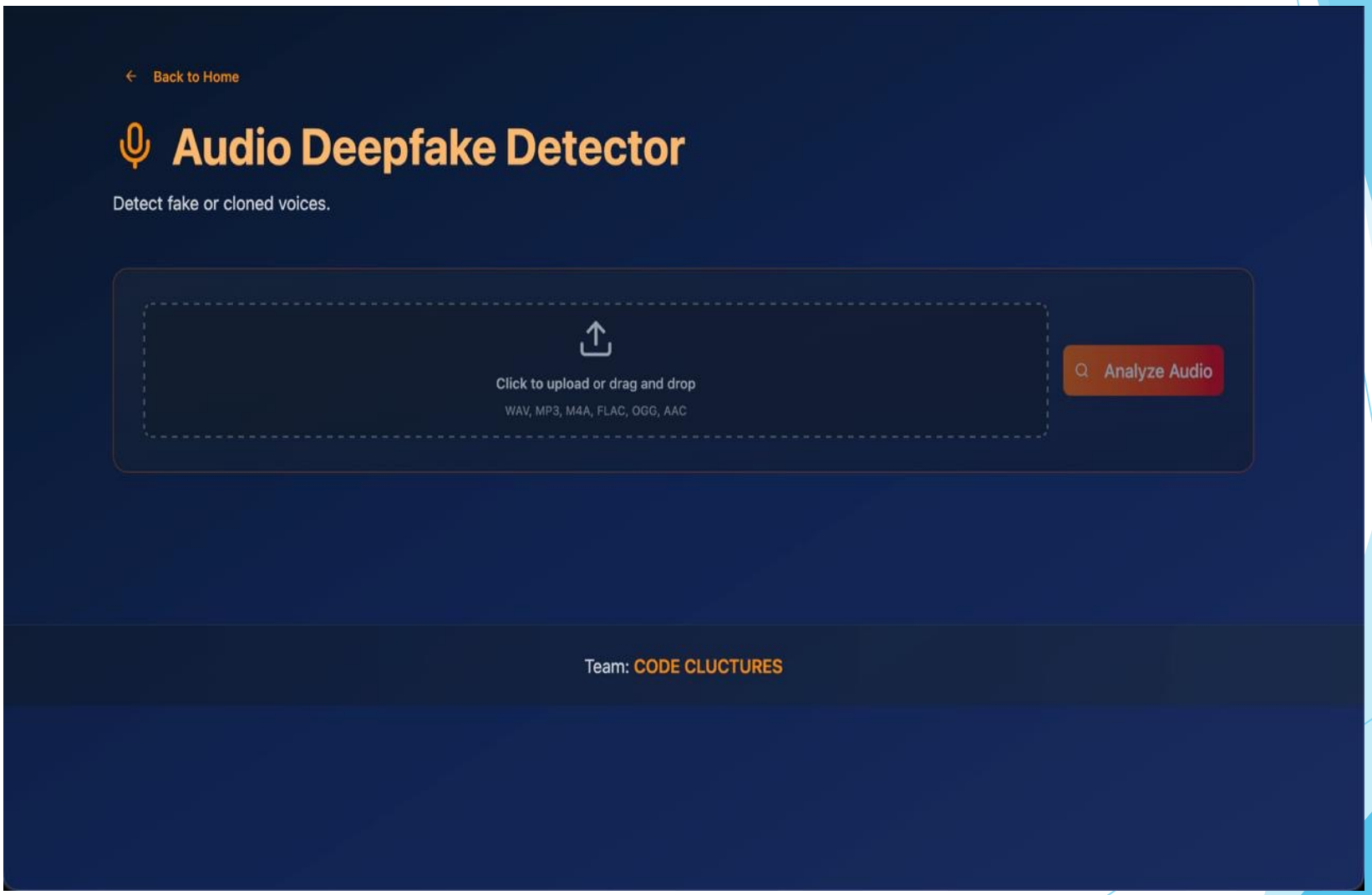

Conclusion

- **Successful Implementation:** Developed an advanced AI-driven system, **BitTruth**, capable of accurately detecting deepfakes in both images and videos.
- **Effective Multi-Model Approach:** By integrating architectures like XceptionNet and Vision Transformers, the system achieves high reliability in identifying subtle facial inconsistencies.
- **Enhanced Digital Trust:** The project provides a practical solution to combat misinformation and fraud by offering users a clear confidence score and authenticity verdict.
- **Future Scalability:** The current framework lays a strong foundation for real-time verification and can be further expanded to adapt to evolving digital manipulation techniques.

References

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Thank You...!!